

Programming assignment

1. (a) by Assignment 6 - Classification

(b) GDA has LDA & QDA,

we know the shape of Taiwan probably looks like ellipse (compared to a line) $\Rightarrow$ we use QDA

Suppose $p(x|y=k) \sim N(\mu_k, \Sigma)$

by the maximal likelihood function, calculate μ_k, Σ_k

In our code $k = 0, 1$, $\vec{x} = (\text{經度}, \text{緯度})$
 $\uparrow \quad \uparrow$
-999 有溫度

by Bayes's rule $\Rightarrow P(Y=1|x) = \frac{P(x|Y=1)P(Y=1)}{P(x|Y=1)P(Y=1) + P(x|Y=0)P(Y=0)}$

and the decision boundary

is the collect of x which $P(Y=0|x) = P(Y=1|x)$

if $\Sigma_0 = \Sigma_1 \Rightarrow$ LDA, boundary is a line

$\Sigma_0 \neq \Sigma_1 \Rightarrow$ QDA, boundary is quadratic

(c) result:

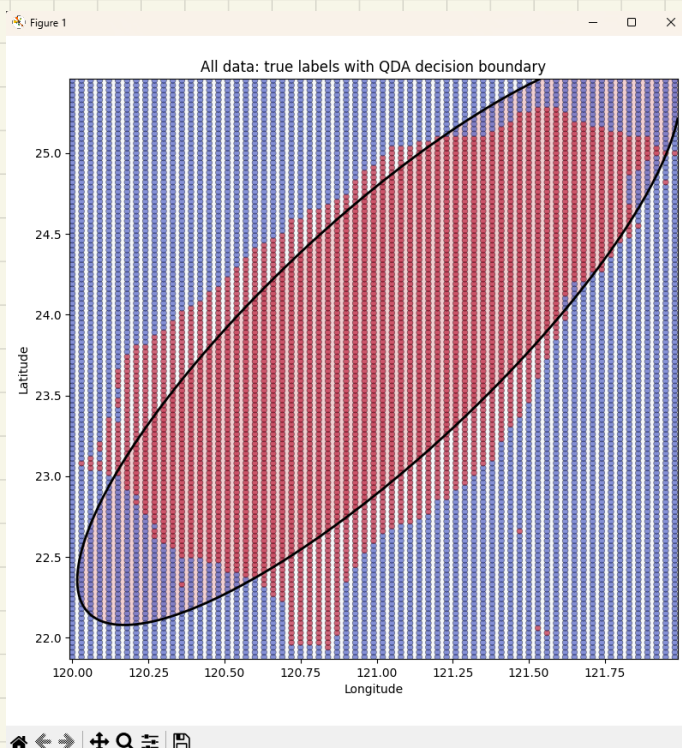
驗證集準確率: 0.8389
測試集準確率: 0.8315

accuracy of test set = 0.8315

It is worse than the random forest
in assignment 4.

The reason is probably about the
boundary of Taiwan is not a perfect
ellipse, and its boundary is not smooth

(d)



2,

(a) by Assignment 6 - Advanced Regression

(b) by the image in 2-(d), we can see the point which is out of boundary, its will not have temperature, so the classification then regression is success.

(c) $C(x) \Rightarrow$ the classification in assignment 4
 $R(x) \Rightarrow$ the regression in assignment 4

$C = C([lon, lat])$

if $C == 1$

return $R([lon, lat])$

else

return -999

(d)

Figure 1

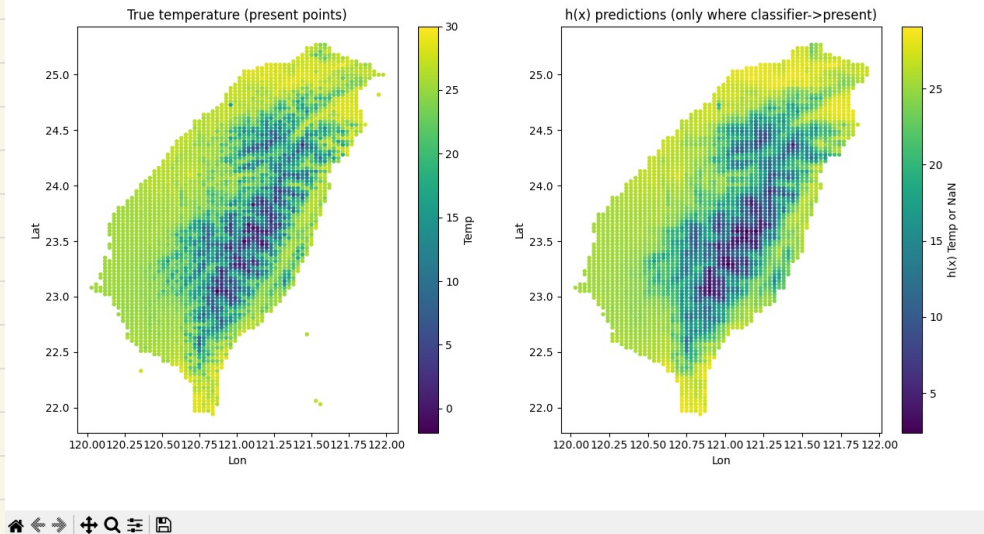


Figure 1

